

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Code of Conduct

I Abinaya Sri J (192524429) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student:

Annexure A – Agile Sprint Planning & User Story Workshop

1. PROJECT VISION AND STAKEHOLDERS

1.1 Project Vision

The vision of the project is to develop a simple, secure, and user-friendly healthcare appointment management system that reduces waiting time and makes appointment booking easier for patients. The system should provide real-time doctor availability and help avoid duplicate or conflicting appointments.

The system will also reduce manual work for receptionists and administrators by maintaining appointment and patient information in a centralized database. Doctors can easily view their schedules and manage appointment status. Overall, the system aims to improve the efficiency and quality of appointment management in hospitals and clinics.

1.2 Project Objectives

The main objectives of the project are:

- To provide online patient registration and login.
- To allow patients to search for doctors by department and availability.
- To display available appointment slots.
- To allow patients to book appointments online.
- To provide cancellation and rescheduling facilities.
- To reduce long queues and waiting time.
- To avoid duplicate and conflicting appointments.
- To allow doctors to view and manage their schedules.
- To provide appointment reminders and notifications.
- To help administrators manage doctors, patients, departments, and appointments.
- To maintain healthcare appointment information securely.
- To provide useful reports for hospital management.

1.3 Key Stakeholders

The major stakeholders involved in the project are:

Stakeholder	Role
Patients	Register, search doctors, book appointments, cancel or reschedule appointments
Doctors	View schedules and manage appointment status
Receptionists	Register patients and manage appointments when required
Hospital Administrator	Manage doctors, patients, departments, slots, and system activities
Hospital Management	View reports and monitor overall system performance
Development Team	Design, develop, test, and maintain the software

System Administrator	Manage server, access control, security, and system configuration
Database Administrator	Maintain database, backup, and data security
Testing Team	Test system functionality, performance, and reliability

2. EPICS AND USER STORIES

The major requirements of the healthcare system are divided into different **Epics**. Each Epic represents a large feature area of the system.

Epic 1 – Patient Registration and Login

User Story 1.1

As a patient, I want to create an account, so that I can use the healthcare appointment system.

User Story 1.2

As a patient, I want to log in securely, so that I can access my personal information and appointments.

User Story 1.3

As a patient, I want to update my profile, so that my personal information remains correct.

Epic 2 – Doctor Search and Availability

User Story 2.1

As a patient, I want to search for doctors by department, so that I can find the appropriate doctor.

User Story 2.2

As a patient, I want to view doctor availability, so that I can select a suitable appointment slot.

User Story 2.3

As a doctor, I want to update my available time slots, so that patients can see my current availability.

Epic 3 – Appointment Management

User Story 3.1

As a patient, I want to book an available appointment slot, so that I can visit the doctor at a suitable time.

User Story 3.2

As a patient, I want to cancel an appointment, so that the slot can become available for another patient.

User Story 3.3

As a patient, I want to reschedule an appointment, so that I can choose another suitable time.

User Story 3.4

As a doctor, I want to view my appointment schedule, so that I can prepare for my patients.

Epic 4 – Notification Management

User Story 4.1

As a patient, I want to receive appointment confirmation, so that I know my booking was successful.

User Story 4.2

As a patient, I want to receive appointment reminders, so that I do not forget my scheduled appointment.

User Story 4.3

As a doctor, I want to receive appointment updates, so that I am aware of changes in my schedule.

Epic 5 – Administration

User Story 5.1

As an administrator, I want to add and manage doctors, so that the doctor information remains updated.

User Story 5.2

As an administrator, I want to manage departments and appointment slots, so that the system remains organized.

User Story 5.3

As an administrator, I want to view appointment reports, so that I can monitor hospital activities.

3. ACCEPTANCE CRITERIA

Acceptance criteria define the conditions that must be satisfied for a user story to be considered complete. The **Given–When–Then** format is used below.

User Story 1.1 – Patient Registration

User Story:

As a patient, I want to create an account, so that I can use the healthcare appointment system.

Acceptance Criteria:

- **Given** the patient is on the registration page,
- **When** the patient enters valid required details and submits the form,
- **Then** the system should create the patient account successfully.
- **Given** the patient enters an already registered email or phone number,
- **When** the registration form is submitted,
- **Then** the system should display an appropriate error message.

User Story 2.1 – Search Doctor

User Story:

As a patient, I want to search for doctors by department, so that I can find the appropriate doctor.

Acceptance Criteria:

- **Given** the patient is logged into the system,
- **When** the patient selects a department,
- **Then** the system should display doctors belonging to that department.
- **Given** doctors are available in the selected department,
- **When** the search is completed,
- **Then** the system should display doctor names and relevant availability information.

User Story 2.2 – View Doctor Availability

User Story:

As a patient, I want to view doctor availability, so that I can select a suitable appointment slot.

Acceptance Criteria:

- **Given** a doctor has available slots,
- **When** the patient selects the doctor,
- **Then** the system should display the available appointment slots.
- **Given** there are no available slots,
- **When** the patient checks the doctor's schedule,
- **Then** the system should display a suitable message indicating that no slots are available.

User Story 3.1 – Book Appointment

User Story:

As a patient, I want to book an available appointment slot, so that I can visit the doctor at a suitable time.

Acceptance Criteria:

- **Given** the patient is logged in and an appointment slot is available,
- **When** the patient selects the slot and confirms the booking,
- **Then** the system should create the appointment successfully.
- **Given** an appointment slot has already been booked,
- **When** another patient tries to select the same slot,
- **Then** the system should prevent duplicate booking.
- **Given** the appointment is successfully booked,
- **When** the booking process is completed,
- **Then** the system should display the appointment confirmation.

User Story 3.2 – Cancel Appointment

User Story:

As a patient, I want to cancel an appointment, so that the slot can become available for another patient.

Acceptance Criteria:

- **Given** the patient has an upcoming appointment,
- **When** the patient selects the cancel option,
- **Then** the system should cancel the appointment.
- **Given** the appointment is cancelled,
- **When** the cancellation is completed,
- **Then** the appointment slot should become available according to the system rules.

User Story 3.3 – Reschedule Appointment

User Story:

As a patient, I want to reschedule an appointment, so that I can choose another suitable time.

Acceptance Criteria:

- **Given** the patient has an existing appointment,
- **When** the patient selects reschedule,
- **Then** the system should display available alternative slots.
- **Given** the patient selects a new available slot,
- **When** the patient confirms the change,

- **Then** the system should update the appointment with the new date and time.

User Story 3.4 – Doctor Schedule

User Story:

As a doctor, I want to view my appointment schedule, so that I can prepare for my patients.

Acceptance Criteria:

- **Given** the doctor is logged into the system,
- **When** the doctor opens the appointment schedule,
- **Then** the system should display the doctor's upcoming appointments.
- **Given** an appointment is cancelled or rescheduled,
- **When** the doctor views the schedule,
- **Then** the updated appointment information should be displayed.

User Story 4.2 – Appointment Reminder

User Story:

As a patient, I want to receive appointment reminders, so that I do not forget my scheduled appointment.

Acceptance Criteria:

- **Given** the patient has a confirmed appointment,
- **When** the scheduled reminder time is reached,
- **Then** the system should send an appointment reminder through the configured notification service.

4. PRODUCT BACKLOG PRIORITIZATION

The product backlog contains all the required user stories. The stories are prioritized based on **business value, patient needs, dependencies, and implementation importance**.

Priority Levels

- **High (P1):** Required for the basic working system.
- **Medium (P2):** Important features that improve the system.
- **Low (P3):** Additional features that can be implemented later.

ID	Epic	User Story	Priority
US1.1	Patient Management	Patient registration	P1 – High
US1.2	Patient Management	Patient login	P1 – High
US2.1	Doctor Management	Search doctor by department	P1 – High

US2.2	Doctor Availability	View available slots	P1 – High
US3.1	Appointment Management	Book appointment	P1 – High
US3.4	Appointment Management	Doctor views schedule	P1 – High
US3.2	Appointment Management	Cancel appointment	P2 – Medium
US3.3	Appointment Management	Reschedule appointment	P2 – Medium
US4.1	Notification	Appointment confirmation	P2 – Medium
US4.2	Notification	Appointment reminder	P2 – Medium
US1.3	Patient Management	Update patient profile	P2 – Medium
US2.3	Doctor Availability	Doctor updates availability	P2 – Medium
US5.1	Administration	Manage doctors	P2 – Medium
US5.2	Administration	Manage departments and slots	P2 – Medium
US5.3	Administration	View appointment reports	P3 – Low

Reason for Prioritization

Patient registration and login are given the highest priority because users need secure access before using other services. Doctor search and availability are also high priority because they are required before booking an appointment.

Appointment booking is the main function of the system, so it is included in the first priority group. Cancellation, rescheduling, notifications, administration, and reporting can be developed after the basic booking process is working.

5. SPRINT BACKLOG – SPRINT 1

The first sprint will focus on developing the basic functionality required for patients to register, log in, search for doctors, view available slots, and book an appointment.

Sprint 1 Selected User Stories

User Story	Description	Priority
US1.1	Patient registration	High
US1.2	Patient login	High
US2.1	Search doctor by department	High
US2.2	View doctor availability	High
US3.1	Book appointment	High

Sprint 1 Task Breakdown

US1.1 – Patient Registration

Tasks:

1. Design registration form.
2. Create patient database table.
3. Implement registration logic.
4. Validate patient details.
5. Check duplicate email/phone.
6. Test registration functionality.

US1.2 – Patient Login

Tasks:

1. Design login page.
2. Implement authentication.
3. Validate username/email and password.
4. Implement secure session handling.
5. Display login error messages.
6. Test successful and unsuccessful login.

US2.1 – Search Doctor

Tasks:

1. Create doctor database structure.
2. Add doctor and department information.
3. Create doctor search interface.
4. Implement department-based search.
5. Display doctor details.
6. Test doctor search.

US2.2 – View Doctor Availability

Tasks:

1. Create doctor availability table.
2. Add available date and time slots.
3. Display available slots.
4. Hide unavailable/booked slots.
5. Test slot display.
6. Verify updated availability.

US3.1 – Book Appointment

Tasks:

1. Create appointment database table.
2. Create appointment booking page.
3. Connect available slots with booking.
4. Validate selected slot.
5. Prevent duplicate booking.

6. Store appointment details.
7. Display booking confirmation.
8. Test appointment booking.

6. EFFORT ESTIMATION USING STORY POINTS

Story points are used to estimate the relative effort, complexity, and uncertainty involved in completing each user story. The **Fibonacci sequence** is used for estimation.

The values used are:

1, 2, 3, 5, 8, 13

Higher points indicate greater complexity or effort.

Sprint 1 Story Point Estimation

User Story	Story	Story Points	Justification
US1.1	Patient registration	3	Basic form, validation, and database storage are required
US1.2	Patient login	3	Requires authentication and session handling
US2.1	Search doctor	3	Requires doctor and department data with search functionality
US2.2	View doctor availability	5	Requires date/time slot management and availability checking
US3.1	Book appointment	8	More complex because it involves slots, database updates, validation, and duplicate booking prevention

Total Story Points

$3 + 3 + 3 + 5 + 8 = 22$ Story Points

Justification of Estimation

Patient registration and login are relatively simple features, so they are estimated at 3 points each. Doctor search also has limited complexity because it mainly involves searching stored doctor information.

Doctor availability is estimated at 5 points because the system must handle dates, time slots, and availability. Appointment booking is estimated at 8 points because it depends on doctor availability and requires validation, database updates, confirmation, and prevention of duplicate bookings.

7. SPRINT GOAL AND EXPECTED DELIVERABLES

7.1 Sprint Goal

“To develop the basic patient appointment booking functionality by implementing patient registration, secure login, doctor search, doctor availability, and appointment booking.”

The main objective of Sprint 1 is to create a basic working version of the healthcare appointment system. At the end of the sprint, a patient should be able to create an account, log in, search for a doctor, view available slots, and book an appointment.

7.2 Expected Sprint Deliverables

At the end of Sprint 1, the following features are expected to be completed:

- Patient registration module.
- Secure patient login module.
- Doctor and department search.
- Doctor availability display.
- Appointment slot selection.
- Appointment booking functionality.
- Basic database for patients, doctors, availability, and appointments.
- Duplicate appointment prevention.
- Appointment confirmation.
- Basic testing of all Sprint 1 features.

7.3 Definition of Done

A user story will be considered complete when:

- The required functionality is implemented.
- The feature works according to the user story.
- Required data is stored correctly.
- Basic validation is completed.
- The feature is tested successfully.
- Major errors are fixed.
- The feature is ready to demonstrate to stakeholders.

8. SPRINT PLANNING SUMMARY

The Agile approach allows the healthcare system to be developed in small and manageable parts. Instead of developing the entire system at once, the project is divided into multiple sprints.

Proposed Sprint Plan

Sprint	Main Focus	Expected Features
Sprint 1	Basic Appointment Booking	Registration, Login, Doctor Search, Availability, Booking

Sprint 2	Appointment Management	Cancellation, Rescheduling, Doctor Schedule
Sprint 3	Notifications & Administration	Confirmation, Reminders, Doctor/Department Management
Sprint 4	Reports & Improvements	Reports, testing, performance, security improvements

This sprint-based approach allows the development team to receive feedback after every sprint. Problems can be identified early and improvements can be added in the next sprint.

9. EXPECTED BENEFITS OF AGILE SPRINT PLANNING

The Agile sprint planning approach provides several benefits for the healthcare appointment system:

- The system can be developed step by step.
- High-priority patient requirements are implemented first.
- Stakeholders can provide feedback after each sprint.
- Errors can be identified and corrected early.
- Changes in requirements can be handled more easily.
- The development team can clearly divide the work.
- Story points help estimate the effort required.
- The product backlog provides a clear list of pending requirements.
- Sprint goals help the team remain focused.
- Working software can be delivered at the end of every sprint.

10. CONCLUSION

The Agile Sprint Planning and User Story Workshop provides a structured way to develop the Smart Healthcare Appointment and Patient Management System. The project vision identifies the main purpose and stakeholders, while Epics and User Stories divide the requirements into smaller and understandable features.

Acceptance criteria help the development and testing teams understand when a feature is complete. The prioritized product backlog identifies the most important requirements, and the Sprint 1 backlog focuses on the basic appointment booking process. Story point estimation helps the team understand the effort required for each user story.

The selected Sprint 1 goal is to provide the basic patient appointment booking functionality. Further sprints can add cancellation, rescheduling, notifications, administration, reports, and other improvements. This approach makes the project easier to develop, test, maintain, and improve based on stakeholder feedback.

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25